

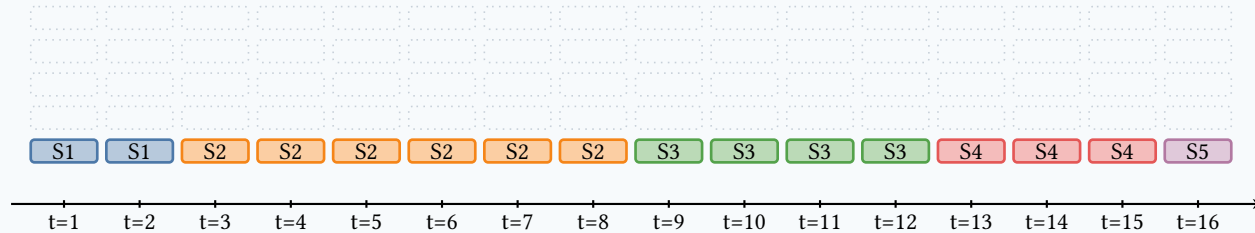
GPU Batching Strategies for Atomistic Simulations

Ten structures needing 2, 6, 4, 3, 6, 5, 2, 4, 3, 2 ionic steps, on 5 GPU slots

Unbatched Simulations

20% slots busy

37 steps to finish all 10

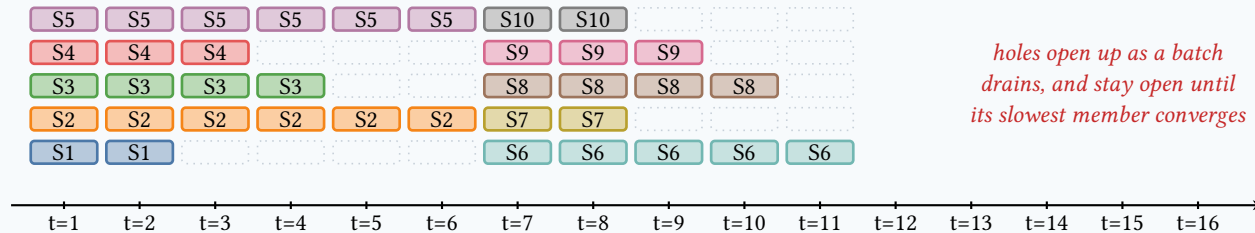


...21 more steps

Binning AutoBatcher

67% slots busy

11 steps to finish all 10

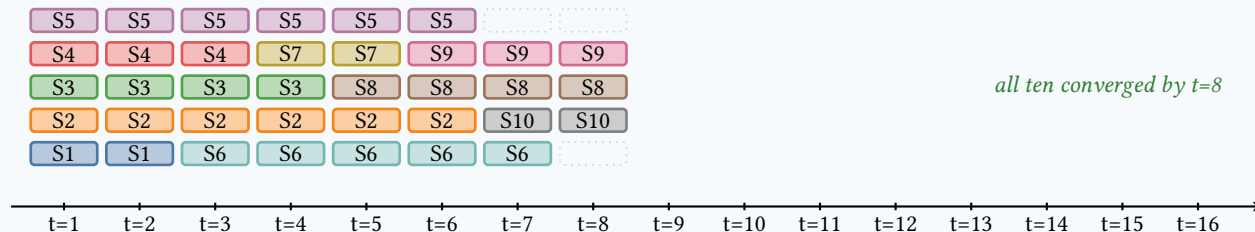


holes open up as a batch drains, and stay open until its slowest member converges

InFlight AutoBatcher

93% slots busy

8 steps to finish all 10



all ten converged by t=8

Unbatched runs one structure at a time, leaving four of five slots idle. **Binning** fills a batch once and holds every slot until the batch's slowest member converges, so the holes grow as the batch drains. **In-flight** refills a slot the moment its structure converges — the color and label changing mid-row is a new structure taking over — which finishes the same work in 8 steps instead of 11.